

Human Factors Research Portfolio 2010 - 2026

Xiao Ge, PhD

Qualitative research to inform
early-stage product
innovation

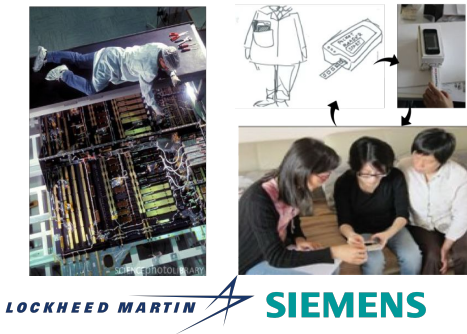
Mixed-methods research on
technology-mediated
interactions

Human factors research on the
smart wearables

Stanford + Siemens
2010 - 2015

Stanford
PhD (2016 - 22) + Postdoc (22 - 24)

Google
2025 -

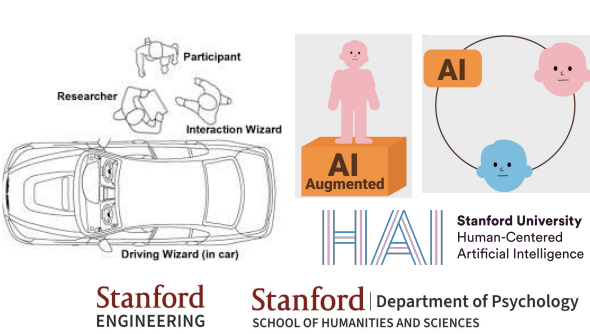


Role:

Need finder	Design engineer
	Trainer and coach

Research focus:

Contextual inquiry	Iterative UX research
Persona building	Qualitative data analysis
Focus groups	



Role:

PhD student	Postdoc fellow	Lead researcher
		Lecturer

Research focus:

Study design	Naturalistic field study	1:1 interview
Survey		Analysis and reporting
Usability study	Interaction analysis	



Role:

Human Factors Researcher

Research focus:

Study design	Methods development	1:1 interview
Lab Study		Survey
Ergonomics research	Analysis and reporting	Design HF evaluation

2010-2015

Qualitative User Research for Early-Stage Product Design and Innovation

Need finder

Design engineer

Trainer and coach

Contextual inquiry

Iterative UX research

Persona building

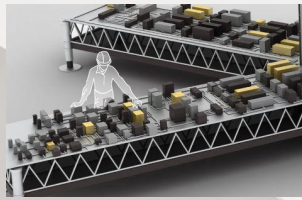
Qualitative data analysis

Impact on innovation strategy, product design direction / product launch, more research & development funding,

Graduate student

2010-2012

LOCKHEED MARTIN



Transformed satellite assembly ergonomics

Design Thinking trainer, coach and lead

2012-2014

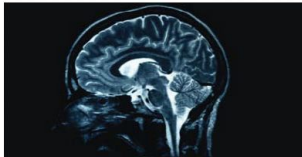
SIEMENS



Enhanced mine security with wireless lifesaver



Market research of the future of gas turbine in China



Personalized brain stroke care with augmented decision-making system



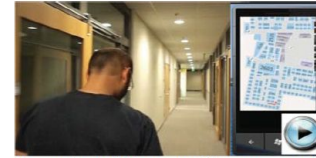
Intelligent traffic management with human as smart sensors

d. HASSO PLATTNER
Institute of Design at Stanford



Amy Robe & 11 months old Connor	Erica Werner & 7 months old Myrabelle	Allison Williams & 11 months old Naomi	Ginnie Pollock & 2 months old Madison	Hannah Kim & 9 months old Alex
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Convenient healthy eating for new moms



Interactive building responding to user needs - indoor navigation



Design of natural while appetizing lighting in fashion shops



Distributed smart motor system for small and median enterprises



Accurate and affordable urine analysis for small hospitals



Boost of innovation with Intellectual Property counseling services



Simple and safe gasification system



Interactive building responding to user needs - building control



Effective operation and maintenance of airport baggage handling by social networking

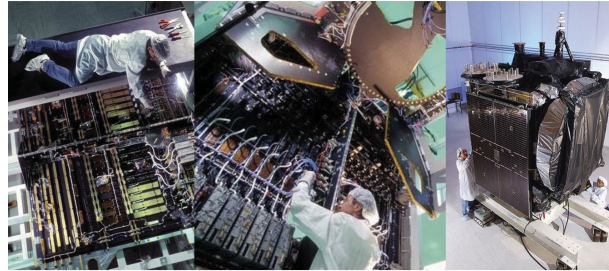
Request: Make satellites more affordable, producible, testable, scalable and modular



Research methods:

Contextual inquiry

Expert interview



(Pictures are used for illustration. No pictures were allowed in the Lockheed Martin facility)

Persona building



User requirement and specification development

Specification	Metric	Rationale
Improves box access	Up until the point when assembly is "locked" and full system tests begin, any box can be accessed without cutting a hole in the structure or requiring users to assume ergonomically undesirable positions.	Currently 75% of satellites suffer from lack of access late in the assembly, which can cost up to 4 months & millions of dollars. Boxes are rarely replaced after full system tests begin.

Outcome and impact:

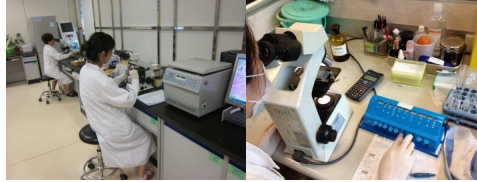
- My 2010-11 project became a classic case study for the class
- Project outcome was adopted by Lockheed, saving approximately \$150 million per year and \$20 million per satellite according to Lockheed Martin liaison Eric Byler (posted in Forbes)

Request: Develop €50 urine strip reader for Chinese Market

Research methods:

Contextual inquiry

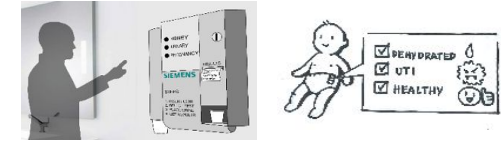
1:1 interview



Focus groups



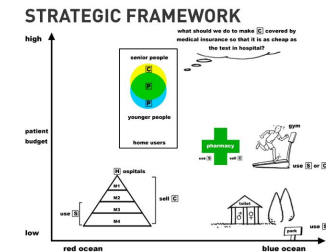
Storyboards



Role playing



Data analysis and synthesis



Outcome and impact:

- Deeper and broader view of unmet needs in China + strategic opportunity areas of product innovation
- Strong support from key stakeholders (e.g., the Head of digital experience, point-of-care at BU) + Secured more research funding

[During PhD]

“Ghost driver”: Implicit interaction at the crossroads

Research assistant

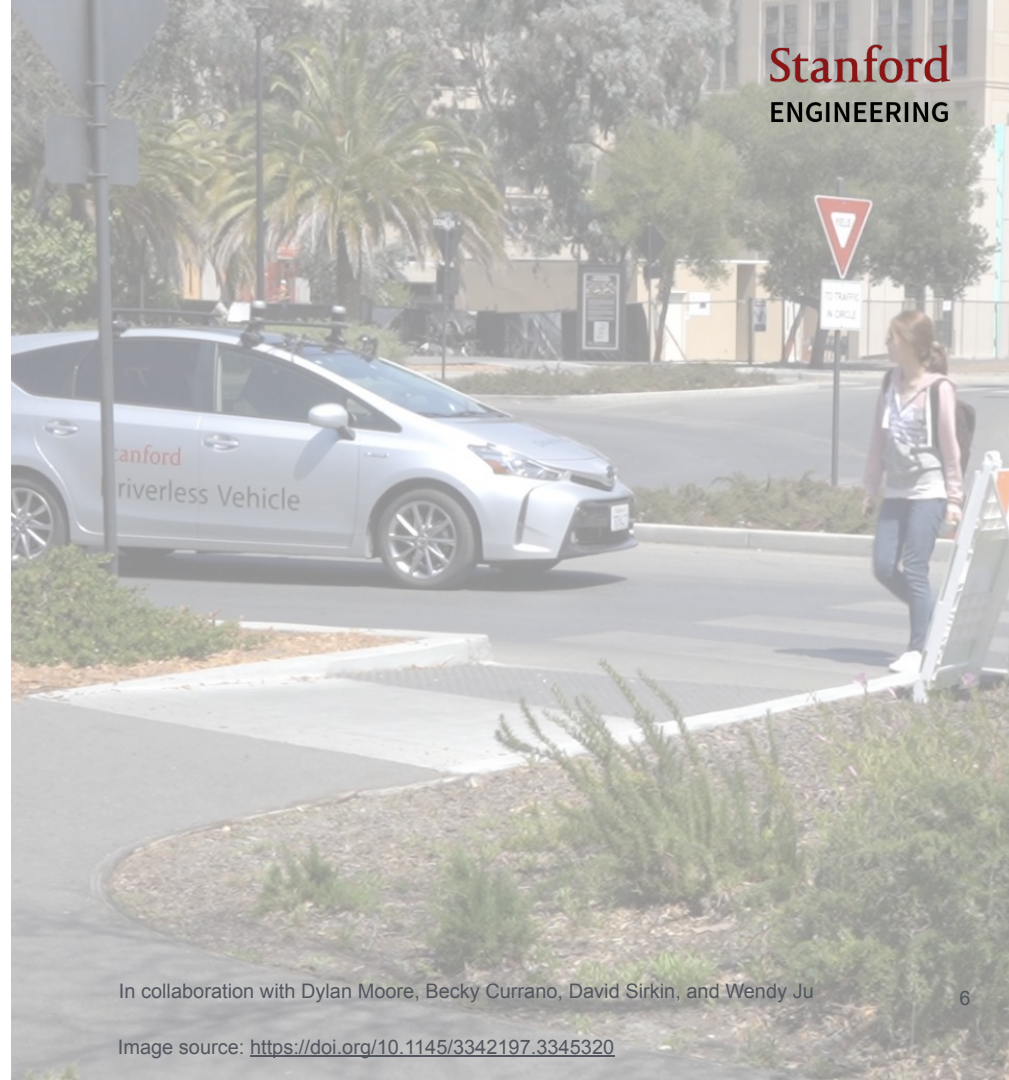
1:1 interview

Study execution

Video analysis

- Conducted interviews, surveys with pedestrians (users) that interacted with the “driverless” cars;
- Performed open-ended coding of video data
- Came up with car behavior scheme

Potential impact on autonomous car behavior design



In collaboration with Dylan Moore, Becky Currano, David Sirkin, and Wendy Ju

Image source: <https://doi.org/10.1145/3342197.3345320>

[During PhD]

Multivariate modeling of intra-individual variability based on time-series observational data

Lead researcher

Study design

1:1 interview

Analysis and reporting

Naturalistic field study

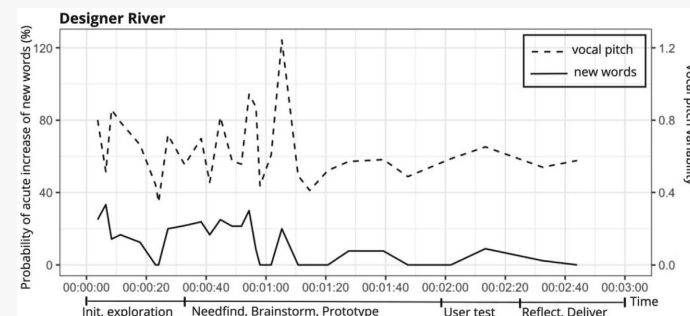
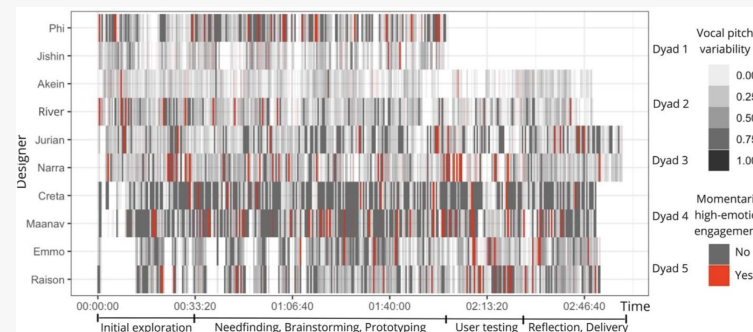


Figure 1. Examples of types of time-series analysis and data visualization I can perform

[Postdoc]

How Culture Shapes What People Want from AI

Lead researcher

Survey

Study design

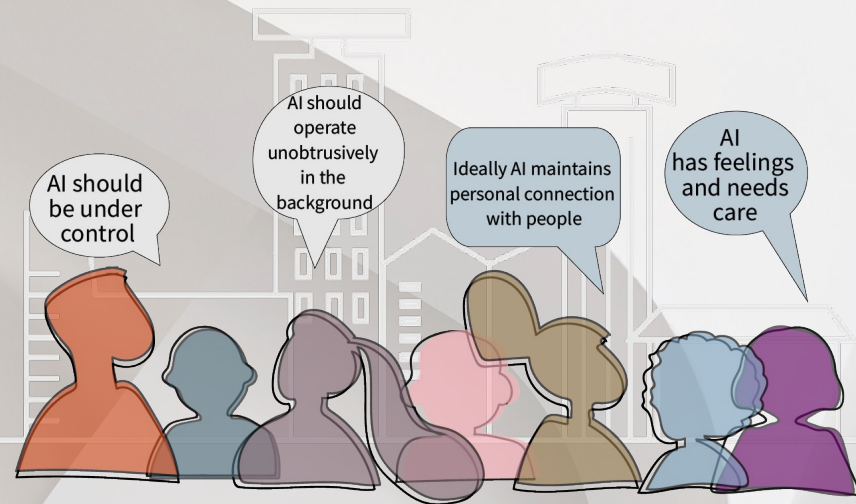
Analysis and reporting

Potential impact on various AI-based products

Stanford School of Humanities and Sciences



Stanford University
Human-Centered
Artificial Intelligence



In collaboration with Chunchen Xu, Hazel Rose Markus, Jeanne L. Tsai, Daigo Misaki and Stanford Cultural Collab.

<https://doi.org/10.1145/3613904.364266>

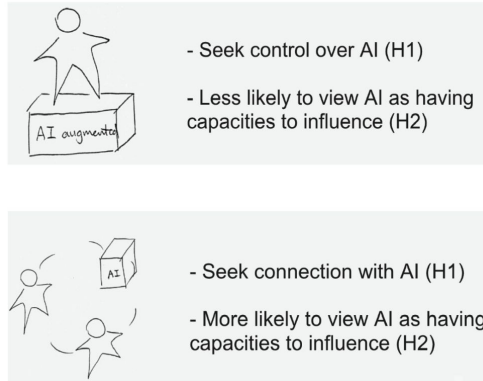
Motivation:

What conceptions of human do we have when we talk about human-centered AI?

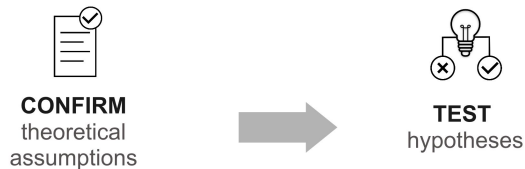
Theory:

Cultural models in self-construal (Markus & Kitayama, 1991)

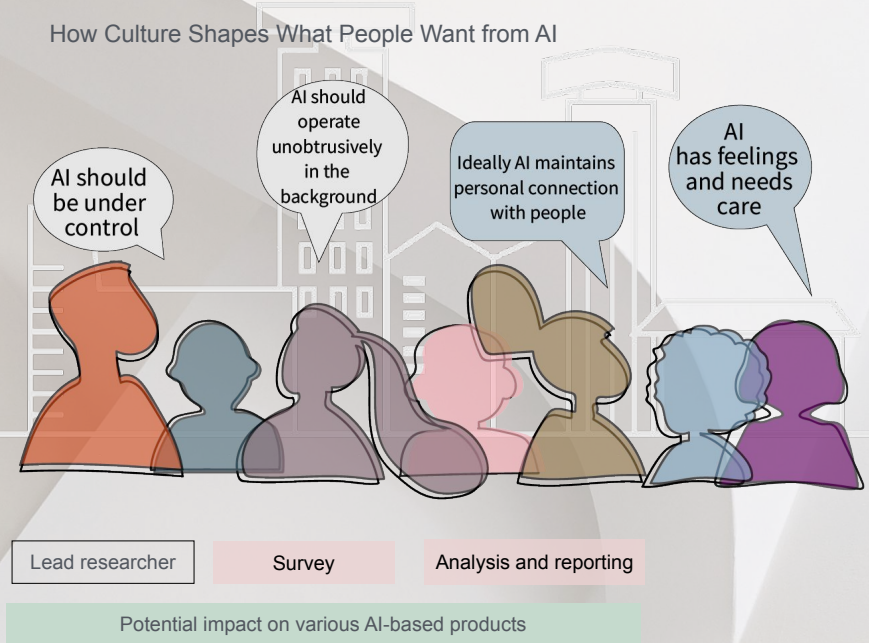
Hypotheses:



Survey study:

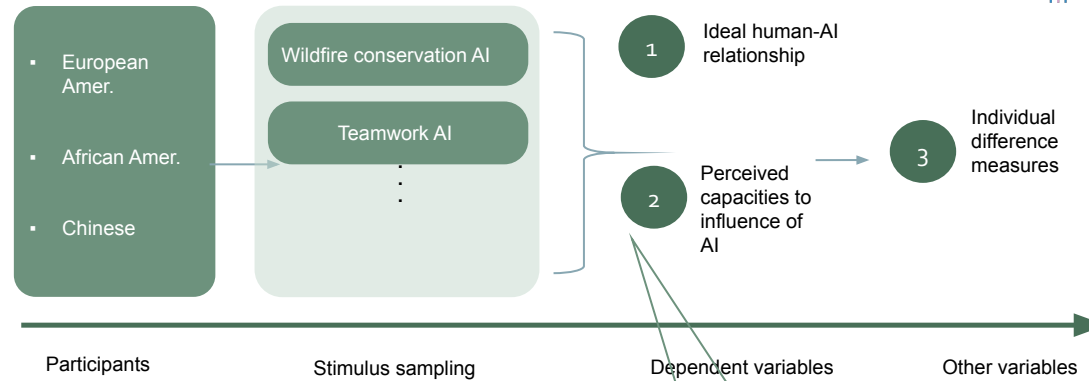


How Culture Shapes What People Want from AI



In collaboration with Chunchen Xu, Hazel Rose Markus, Jeanne L. Tsai, Daigo Misaki and Stanford Cultural Collab.

Main Study design:



Two-option nine-item measure about ideal AI's capacities to influence ($\alpha = .86$)

Low capacities to influence (=1)	High capacities to influence (=2)
AI provides care to but does not need care from people.	AI provides care to but also needs care from people.
AI does not have feelings and emotions.	AI has feelings and emotions.
AI remains an impersonal algorithm to perform tasks.	AI maintains a personal connection with people.
AI operates unobtrusively in the background.	AI participates in social situations.
AI remains as an abstract algorithm whenever possible.	AI has a tangible representation of its existence (e.g., a physical body) whenever possible.
AI has little autonomy.	AI has autonomy.
AI mainly preforms tasks that are pre-planned by humans and has little spontaneity.	AI has spontaneity when performing tasks.
AI behaves consistently across different situations.	AI behaves differently across different situations.
AI interacts with people on terms made by people.	AI interacts with people on terms made by AI.

Findings:

Compared to European Americans*, Chinese prioritize connecting with AI over controlling it, preferring AI to have greater autonomy and emotional capacity. European Americans prefer strict control and minimal AI influence. African Americans align with European Americans on wanting control, but fall between the two groups in their desire for connection and AI autonomy.

*these cultural groupings are not essentialized or monolithic categories.

Findings:

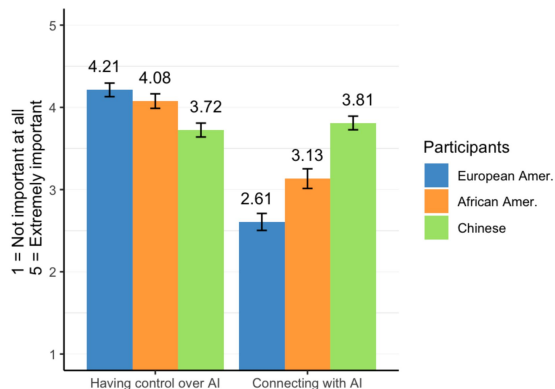


Figure 5: Importance of having control over AI and connecting with AI in Main Study, based on a 5-pt. scale. Error bars represent standard errors.

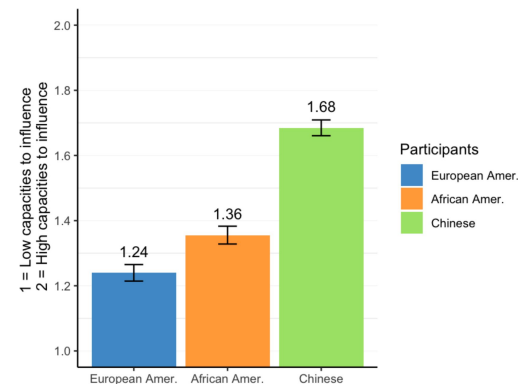


Figure 6: Preferred capacities to influence in ideal AI in Main Study, based on a 2-pt. scale. Error bars represent standard errors.

Impact:

For product design and engineering:

- Rethinking control-based relationships in designing human-AI interaction.
- Broadening the space of the imaginary for conceptualizing AI's characteristics.

For research:

- Empirical approaches to examine people's culturally-shaped preferences regarding AI
- Illuminating the implicit and latent cultural assumptions about humans that are built into current models of human-computer interaction, and through this,
- Expanding current models of human-computer interaction to increase the potential of future technologies.

[Post PhD]

Diagnosis of team performance in immersive VR

Lead data analyst

Experimental
design

Mixed-methods
analysis

Potential impact on VR/XR/AR product design
directions

Stanford
ENGINEERING



Hasso
Plattner
Institut



VHIL
Virtual Human Interaction Lab



In collaboration with Ade Mabogunje, Neeraj Sonalkar, Mark Miller and Jeremy Bailenson

Image source:

<https://www.vjar360.com/how-virtual-reality-is-revolutionizing-remote-teamwork/>

Motivation:

VR
scenario

Formal

VS

Informal

Task content

Divergent ideation

VS

Convergent
decision making

Hypotheses:

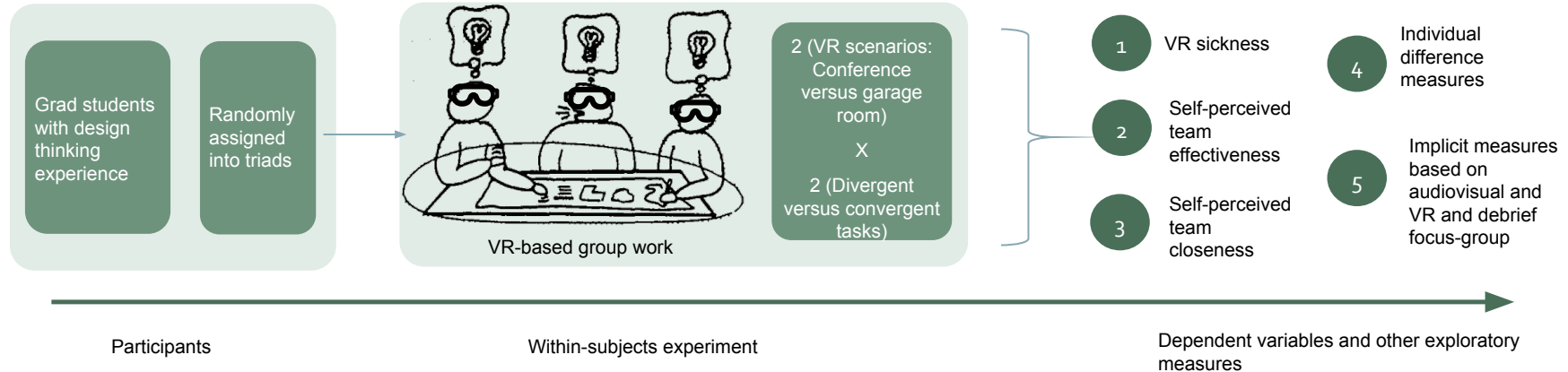
H1: Users will report different levels of VR sickness working in different VR scenarios.

H2: More VR sickness will be perceived in tasks incongruent with the VR work environment.

H3: Self-reported team effectiveness will suffer with VR sickness.

(Other RQs)

Study design:



(Visual built upon Tang, 1989)

Findings:

H1 receives no support.
H2 receives partial support.

H3 receives support— effect of VR sickness level on self-perceived team effectiveness persists even after controlling for team closeness (adj. R-squared = 0.30, $F(2, 96) = 21.72$, $p < .001$, where VR sickness: $B = -0.1$, $t(96) = -2.74$, $p < .001$)

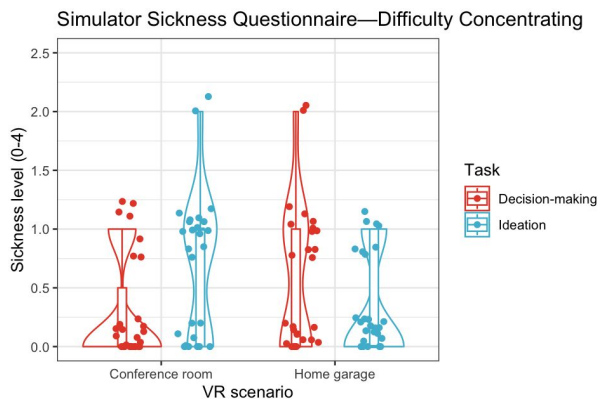


Figure 1. Distribution of SSQ—“difficulty concentrating” by VR scenario and task content, displaying median and interquartile ranges. Data here includes 10 triads of 30 participants in the 4 randomly-ordered session study. Significant interaction effect is found based on negative binomial regression and Wilcoxon signed rank test.

Participant's 1st person perspective: “Unenthusiastic brainstorm” in the conference room



Figure 2. Example here shows near the end of an ideation session in the conference room. The office building-based conference room environment was reported as “distractive”, “unenthusiastic”, etc for brainstorming

Impact:

- User work performance could suffer with even just slight VR sickness from 10-min VR-based work
- Offering insights how to design digital experiences that are conducive to users' work
 - VR/AR/XR as more than “escape or a place for work”

Human factors research on the smart wearables

Human factors researcher

Study design

Methods development

Lab Study

Survey

Ergonomics research

Design HF evaluation

1:1 interview

Analysis and reporting

Direct Impact on product design directions with research evidence

Watch Band Sizing

Watch Comfort, Fit & Stability

Button Ergonomics

Display Readability/ Glanceability

IMU based Watch Stability

ECG Contact Evaluation

Watch Don & Doff

Accessory Band Comfort & Fit

...

Xiao Ge, PhD

- Google Scholar:
<https://scholar.google.com/citations?user=NSIQBQoAAAAJ&hl>